MAPS OUT OF A QUASI-COHERENT OBJECT IN THE CHAOTIC-TOPOLOGY COMPARISON ARE COMPUTED BY THE GLOBAL SECTIONS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is written out in full: the adjunction step, the application of the comparison computation, and the Leray rewriting of the global sections.

Editorial note. The label lemma-QC-hom-out-of-plus-topology names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. The statement refers to the standing comparison of topologies and is delivered verbatim; the two cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- The comparison of a topology with the chaotic one and its quasi-coherent calculus. Same chapter, the standing lemma identifying the quasi-coherent objects of the comparison as the objects whose pushforward is quasi-coherent. Supplies the hypotheses and the quasi-coherent objects appearing in the statement.
- The corresponding computation without the extra topology. Same chapter, the lemma computing Hom out of a quasi-coherent object as the derived Hom of global sections. Supplies the identification applied after the adjunction.
- The Leray identification for the comparison. Leray, applied as recalled in the proof, giving R Gamma(U, M) = R Gamma(U, R epsilon_* M). Supplies the rewriting of the global sections of M.

Source reference. Original reference 0H0S: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 11173–11173. Prerequisite pointer invoked at proof line(s) 11695; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0GZS: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 11608–11608. Prerequisite pointer invoked at proof line(s) 11679; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:sites-cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Lemma 0.1. *Notation and assumptions as in Lemma 0GZS. Suppose that K is an object of $QC(\mathcal{O})$ and M arbitrary in $D(\mathcal{O}_\tau)$. For every object U of $\mathcal{C}$ we have*

$$\mathrm{Hom}_{D((\mathcal{O}_U)_\tau)}(\epsilon^*K|_U, M|_U) = R\,\mathrm{Hom}_{\mathcal{O}(U)}(R\Gamma(U,K), R\Gamma(U,M))$$

Proof. We have

$$\mathrm{Hom}_{D((\mathcal{O}_U)_\tau)}(\epsilon^*K|_U, M|_U) = \mathrm{Hom}_{D((\mathcal{O}_U)_{\tau'})}(K|_U, R\epsilon_*M|_U)$$

by adjunction. Hence the result by Lemma 0H0S and the fact that

$$R\Gamma(U,M) = R\Gamma(U, R\epsilon_*M)$$

by Leray. □